

- (2 points) Find the missing value in Layer 3 of $\text{net}_D(\cdot)$ and state your justification?
- (5 points) What is the output obtained from each layer of $\text{net}_D(\cdot)$?
- (3 points) What is the learning capacity (LC) and computational complexity (CC) of each layer in $\text{net}_D(\cdot)$?

2. You are provided with two networks

$$\begin{aligned} \text{net}_c(\cdot) &\mapsto (1 : \text{Conv2D})64c3w1s1p \rightarrow (2 : \text{ReLU}) \\ &\rightarrow (3 : \text{Conv2D})64c3w2s1p \rightarrow (4 : \text{ReLU}) \end{aligned} \quad (5)$$

$$\text{net}_d(\cdot) \mapsto (1 : \text{Conv2D})256c3w1s1p \rightarrow (2 : \text{ReLU}) \rightarrow (3 : \text{PixelShuffle})4w4s \quad (6)$$

- (a) (5 points) Scale the depth of $\text{net}_c(\cdot)$ in order to double the number of trainable parameters, and demonstrate it numerically?
- (b) (5 points) Scale the width of $\text{net}_d(\cdot)$ in order to scale up the number of trainable parameters by 1.6 times, and demonstrate it numerically? Please take into account the constraints of operational integrity of the layers.

3. A semantic segmentation network is defined accordingly

$$\begin{aligned} \text{net}_E^{(B)}(\cdot) &\mapsto (1 : \text{Conv2D})16c5w1s0p \rightarrow (2 : \text{ReLU}) \rightarrow (3 : \text{Conv2D})16c5w1s0p \\ &\rightarrow (4 : \text{ReLU}) \rightarrow (5 : \text{MaxPool})4w4s \end{aligned} \quad (7)$$

$$\begin{aligned} \text{net}_D^{(B)}(\cdot) &\mapsto (1 : \text{Conv2D})16c5w1s0p \rightarrow (2 : \text{ReLU}) \rightarrow (3 : \text{Conv2D})16c5w1s0p \\ &\rightarrow (4 : \text{ReLU}) \rightarrow (5 : \text{MaxUnPool})4w4s(\text{net}_E^{(B)} : 5) \end{aligned} \quad (8)$$

$$\text{net}_E(\cdot) = \text{net}_E^{(B)} \left(\text{net}_E^{(B)} \left(\text{net}_E^{(B)}(\cdot) \right) \right) \quad (9)$$

$$\text{net}_D(\cdot) = \text{net}_D^{(B)} \left(\text{net}_D^{(B)} \left(\text{net}_D^{(B)}(\cdot) \right) \right) \quad (10)$$

When provided with an input $\mathbf{x} \in \mathbb{R}^{3 \times M \times N}$, so as to obtain the semantically segmented output $\mathbf{y}$, such that $\mathbf{z} = \text{net}_E(\mathbf{x})$ and $\mathbf{y} = \text{net}_D(\mathbf{z})$

- (a) (5 points) Draw the diagram of the complete network indicating clearly all the data flow?
- (b) (3 points) What should be the minimum value of M and N for this network to function correctly?
- (c) (2 points) What would then be the nature and size of $\mathbf{z}$ and $\mathbf{y}$ for the value of M, N which you have stated above?

Part B

1. (5 x 2)
- a) In a 4 class classification task, the class labels are 1,2,3,4. The ground truth label of a sample is 3. The output produced by a neural net for the sample gives the probabilities (0.6, 0.2, 0.1, 0.1). Compute cross entropy error for this prediction.
 - b) In L2 regularization, what will happen if the weight parameter associated with the regularization term becomes too high?
 - c) Let p be the probability of dropping a neuron during training. At test time, how the contribution of a neuron is determined?
 - d) What is the difference between cross attention and self-attention? Explain with an example.
 - e) All the parameters of a neural network are initialized to large values and then mini-batch SGD is applied to train. What would be the most likely consequence?
2. (2 x 5)
- a) Design a 3 layer RNN that will take an email (in English) and classify it to spam or not spam. The training data will have a set of emails and their labels (spam or not spam). You must mathematically define the hidden states in every layer, the error function and the diagram of the RNN.
 - b) If the training data set is re-scaled by a particular factor, do the learned weights of either batch normalization change? If only a small subset of points in the training data set are re-scaled, how this impacts the learned weights of batch normalization?
3. (2 x 5)
- a) Suppose that you have a large database of biological strings containing sequences of nucleobases drawn from $\{A, C, T, G\}$. For example, AATCCGCTAG is a possible sequence. Some of these strings contain unusual point mutations representing changes in exactly one symbol in the sequence. For example, if the original sequence is AATCCGCTAG, then a mutated sequence can be ATTCCGCTAG. Propose an unsupervised method using LSTM to detect these mutations. You must mathematically define the computation of each component of the network along with the diagram.
 - b) For the above problem, if you have a training dataset where for each sequence the mutation positions are available, how do you modify your LTSM architecture? You must mathematically define the computation of each component of the network along with the diagram.
4. (2 x 5)
- a) In an encoder-decoder architecture what is the role of attention mechanism? How is attention computed (define mathematically along with a diagram)?
 - b) What is the role of multi-headed attention in transformer? How is this computed (define in terms of a function of matrix and vectors)?